WunderLanguage — Design Handover

Live Japanese game-text capture that compounds with every onboarded game.
Full project transfer: mission, rules, architecture, measured history, loops, state, plans.
Dated 2026-10-13. Supersedes chat history; BOOKS.md remains the commit-level ledger.

1. What this document is

This is the full handover for WunderLanguage, a live Japanese-to-English game-translation overlay. It exists so a new chat session can operate the project at full understanding without re-learning months of measured decisions from scratch.

How to use it: treat sections 2-3 as binding operating rules, sections 4-6 as verified architecture and history, sections 7-9 as the compounding core of the project, sections 10-14 as current state and plans, section 15 as the questions only the user can answer. Do not re-litigate shipped verdicts without new evidence. Every behavior batch in this project ships with a revert phrase; they are listed with the batches.

Standing fact about evidence: panel and log proof outrank compile proof. Corpora under %AppData%\WunderLanguage\corpus\ are append-only evidence and are never rewritten. Replay parity and simreplay decide whether behavior changed.

2. Mission and product objective

The mission is a reusable, adaptable system that captures Japanese text displayed in games and publishes faithful Japanese text alongside an English translation. The long-term goal is a defined collection of onboarded games, with the system becoming more capable as each game is added. Shared improvements must strengthen existing games as well as future ones — never a collection of unrelated hand-tuned game implementations.

Three stages, kept distinct. Stage 1 — capture Japanese: detect genuine in-game text, recognize the characters, reconstruct across frames where useful, publish. The objective is every genuine line including brief or one-time appearances; repeated visibility must not be a requirement. Stage 2 — preserve fidelity: published Japanese matches what the game displayed, character for character wherever visual evidence permits. Never substitute plausible Japanese for unreadable characters. Stage 3 — translate: the captured Japanese goes to the translation model; translation quality is evaluated independently and never repairs the source.

Completeness and fidelity are both primary. Missing real dialogue is a failure; publishing unsupported Japanese is also a failure. The system must achieve both rather than optimizing one by sacrificing the other. For an educational tool the release bar is on the order of one misteaching per two hours of play.

3. Governing principles and operating rules

3.1 The trio

- Reliability — capture genuine in-game Japanese as completely and faithfully as possible; keep published text aligned with on-screen text; omission is preferred over misteaching, but omission of difficult text is itself a failure to fix, not a success to keep.
- Scalability — prefer shared, reusable, game-blind capabilities when they solve the underlying problem; new game onboarding should improve or validate shared systems; game-specific rules only on evidence; never improve one game by silently degrading another.
- Adaptability — survive resolution, font, UI, text-speed and setup variance; support transient dialogue, typewriter text, menus, titles and other contexts without forcing all through one rule; degrade gracefully and keep seeking recovery for ambiguous text.

3.2 Evidence discipline

- Measures first, then implementation. One focused change at a time. Every batch ships a revert phrase. Replay parity plus simreplay decide behavior preservation. Panel and log proof, never compile proof alone. Fires alone are never a success criterion — newly recovered Japanese is reviewed against recorded evidence, and merged or incorrect lines count as regressions.

- Corpora never rewritten; add evidence or labels, never rewrite history. Exact-match-only glossary and fixup rules unless a change is supported by evidence. Docs and papers are reference-only: relevance listed before acting, no action until the user approves, no implementation without a cited failure or measured win, no model training or finetuning, no model internals.

3.3 Run protocol

- The user plays; the user never runs commands. Every run request ships exact instructions plus keeper state. Minimize runs — single confirmatory runs. Read simreplay and replay results solo (rapid-sequential launches lose log lines). Red means an old exe is still running: taskkill /F /IM WunderLanguage.Host.exe first, no asking needed.

3.4 Content and roadmap rules

- Profanity blocked only when hallucinated (English profane plus no Japanese alibi — legitimate adult-game content stays). Names use official manual English forms (Mash per manual, reverses Matthew; Wedge/Biggs English still an open user call). Unknown names masked through translation and restored after. Unknown JRPG names (?????? runs) masked through. Shop UI waits on the user's shop-play signal. Text-only communication except geometry or truth shots.

4. System architecture

4.1 Pipeline

- Capture loop (~1s p50): window capture with focus gate (unfocused emulator holds, no OCR) and static-frame skip; OCR via sidecar (EasyOCR detection stock params, manga-ocr recognition, beamed 4 live, 2x upscale, 900px detection cap); stability machine (candidate formation, growth/typewriter tracking, pageturn/flush roads, suspect/stain guards, duplicate and subpart suppression); publish decision kinds (publish-new, growth, flash, pageturn, flush variants, singleton, weak-hold); translation downstream (DeepL primary, CAT-Translate offline fallback, translation cache global and shared).
- Title path: quiet-loop logo detector with hint gating (caps/year/copyright), logo-geometry gate, gameplay veto (no TITLE within 60 loops of dialogue), TITLE-grace prompt window, pack FullTitle triple. Battle path (FFVI): geometry splitter, slots, glossaries, name memo. Menu path: geometry-aware tracking with divergence guards.

4.2 Packs

- Pack = game-scoped data and behavior: signatures (routing by proper-noun votes), title triple, OCR/display fixups, battle/monster/item/weapon vocabs, battle and name glossaries, vision calibration (neutral until cited). Unknown games run universal (degradation baseline: never bank guesses, never post TITLE). Registry routes by ROM-normalized id, window title, and content votes; ties keep current.

4.3 Sidecar and services

- Python sidecar on 127.0.0.1:8000 (leave alone): OCR + CAT translation. 6GB VRAM budget (RTX 3060 laptop class). Survival: job-object ownership, CPU cap with clamp, completion-port detect with 3-attempt auto-restart, low-memory listener, power request, WER keys. Restart budget exhaustion now raises an in-panel engine-dead notice (no popup, no focus steal). Deaths dump uptime, exit code, and ring tail for forensics.

4.4 Evidence and metrics

- Corpus: append-only JSONL (ocr reads with blocks/conf/geometry/beams, decision bubbles with kinds and loop stamps, translation fires). Replay: fixup/normalize path parity (drift must be 0). Simreplay: stability/commit simulation with fire overlap vs recorded, missed/extra sets, [b6metrics] (NE updates-per-fire, lag in loops), [panellat] budgets (p50<=10, p95<=30, max<=60, NE<=0.5), [selfdup] would-block, [spend] translation economics. Keepers: keep-log.armed and keep-shots.armed sentinels (newest 200 kept).

5. Environment and constraints

- Windows, Debug as daily driver, 12 CPU / 32GB reference machine, second own Python server on :8000 left alone, Defender exclusions applied. No model training or finetuning, no model internals, no architectures busting 6GB VRAM. No shop UI until the user signals shop play. Books program completed; papers read for applicable concepts only.

6. History ledger (what was built, with evidence)

6.1 Foundation: survival, performance, vision

- Survival batches live-proven: job ownership with WER keys, 50% CPU hard cap with clamp, completion-port detect with 3-attempt auto-restart, low-memory listener, session power request, job notifications. Perf program closed (14s startup, PERFORMANCE.md SLOs). Vision probes 1-4 plumbed. Revert phrases: revert job+wer, cpu-cap, lowmem, power, job-notify, auto-restart, clamp.
- Zelda repair pack-gated: battle gate, logo hold and window, title purity (marker-row only), fragment hold, turned-page rescue, flush-on-pageturn proven live. Reverts: zelda-battle-gate, zelda-logo-hold, zelda-logo-window, zelda-title-purity, zelda-fragment-hold, zelda-turned-page.

6.2 Bottlenecks 1-9 (offline as far as offline reaches)

- B1 suspect stain and weak-hold lineage. B2 battle vocab (Magitek/Cure/Fire official EN; BattleCommands indices floated). B3 speaker-anchored fixups (rogue-log to Locke forms); box-confidence convicted as presence signal, not quality. B4 scroll-glue guard (>=0.65 on 406 fires). B5 wall-clock pricing (docs only). B6 loop-stamp metrics (simNE/liveNE). B7 recall proxies. B8 cold-start price zero on JA-fire plus the 8-step pack-authoring checklist. B9 panel-latency budget with corrected lag accounting. C3 fuzzy-canonical, A1 averaging trial, sentdedup shipped with verdicts open or banked.

6.3 Bottleneck 10 — first-fire coverage

- Measured: 64-71% of distinct reads are single-glimpse; complete-and-substantial unfired singletons 3/41/46 across three corpora; intro session 166 reads to 2 fires. Confidence-gate autopsy: 0.9 threshold posts hallucinations (1000-yen style) while silencing perfect tutorials — killed as a fix direction.
- Tier 1 fuzzy-window (3551bf3): same-length OCR variants at similarity >=0.85 count as one lineage; first build failed honestly (overlap 76 to 73) and was corrected by preserving successor rescue. Final FFVI-old: 115 to 117 fires, overlap 76=76, missed 7=7, extra 39 to 41, with one genuine clean-form recovery backed by corpus loops 2647-2805. Revert: revert fuzzy-window (kept: revert growth-fast).
- Tier 2 anchored-singleton (90e3b67): button prompts and speaker-start lines fire on first sight under shape guards (terminal, clean start, length 6-30, max one bracket anchor, no ASCII runs). FFVI-new 101 to 104 with +2 distinct genuine and zero removed (worktree-baseline diff); Zelda-long 71 to 72; others unchanged. Blends excluded by the single-anchor precision fix at disclosed recall cost. Revert: revert tier2-anchored.

6.4 Paper Mario onboarding and hardening

- Baseline triage: 1528 loops, 343 reads, 26 fires all from the intro movie (growth-fast carried it); gameplay silence traced to focus holds during screenshotting plus static-frame skips plus a crash — not a pipeline refusal. Sim reproduces 26/26. Pack v1 shipped in checklist order with evidenced-only contents (universal vs pack JA-identical, zero delta — pack value is live-only). ROM-ID normalization (version-proof identity, paren fix included). Title hardening: gameplay veto, TITLE-grace prompt, credits audit (zero Latin-only publications across ~1000 bubbles). Engine-death notice, death forensics, shot session labeling, memo save-on-Stop. Reverts: revert pm-pack, revert rom-id, revert title-harden, revert engine-notice, revert death-forensics, revert shot-labels, revert memo-stop.

7. The five compounding loops (the core that strengthens old games)

Learning here is never model training. It is five symbolic loops — banked pairs, weights, thresholds, vocabs — each measured-first, each game-blind-safe, each with a defined payoff condition. Day-one deltas are mostly zero by design; the value accrues as games onboard.

Loop 1 — global fixup tier (c04f976). Revert: revert loop1-global.

- Mechanism: pairs mined identically (same raw to same fixed) in at least two game files promote to corrections/_global.json at startup; per-game wins on lookup; conflicts never promote; exact-match discipline throughout. Evidence: 70 agreed pairs qualify, 44 conflicts held per-game, FFVI shares zero. Proven live ([miner] global promoted 70/44, matching the offline count exactly). Payoff: the next game gets solved misreads free on day one instead of relearning them.

Loop 2 — confusion re-mining (959fe44). Revert: revert loop2-remine.

- Mechanism: tools/mine-confusion.py re-mines character substitutions from miner files per corpus with counts and source games; reviewer appends qualifying pairs to the hardcoded ConfusionPairs (games>=2, genuine visual confusions; punctuation-normalizer and single-line-cluster pairs excluded by rule). First re-mine: 91 candidates, 29 appended (E/worker, seal-family, voicing, katakana pairs). Blast radius: weighted pass runs only on integer-pass

abstentions. Zero new mappings on all four corpora; replay drift 0. Payoff: future item menus and vocab matching resolve instead of abstaining. Procedure: rerun script per new game, review, gate.

Loop 3 — memo persistence (04ba5d3, closeout 8658118). Reverts: revert loop3-memo, revert memo-stop.

- Mechanism: battle-name memo (short JA to EN, guarded against refusals, echoes, action rows, soup self-lock) persists per pack with a 200-entry cap instead of dying on pack switch; provider-stamped so EN from another provider never loads; save on switch and on Stop. Deliberately no seeded files. Single-game persistence gap caught and fixed pre-closeout; sim hermeticity proven both directions (sim neither writes nor reads memo files). Payoff: second sessions open warm with consistent names. Live proof owed: [memo] saved/loaded lines.

Loop 4 — vocab self-proposal (73fc413). Revert: revert loop4-vocab.

- Mechanism: stable fired JA nominates short terms (2-12 chars) into vocab-proposals/<pack>.json with counts; write-only sidecar on the live publish path after the sim early-return, never read back except by reviewer promotion. Promotion bar: count>=3 or demonstrated vocab gap plus corpus evidence. Zero behavior change by construction (sim cannot create the queue directory — proven). Payoff: pack vocabulary grows from evidence instead of hand-authoring.

Loop 5 — per-game profiles (88ec56b). Revert: revert loop5-profiles.

- Mechanism: four stability-commit knobs (stableReadings 2-4, nearDupe 0.85-0.95, lineageSim 0.55-0.75, fuzzyAdopt 0.80-0.90) read per-pack overrides from profiles/<pack>.json, validated, defaulting otherwise; self-dup and geometry knobs correctly untouched. No profiles exist: zero delta everywhere. Proven both directions with a throwaway probe (stableReadings=2: fires 117 to 145 BUT overlap 76 to 71, missed 7 to 12 — looser fires more and teaches worse; probe deleted, defaults restored exactly). That experiment is the loop's premise priced: knobs matter per game, tuning happens at review with all-corpora re-gate, never by feel. Procedure: simreplay lag/NE per corpus, write profile, re-gate all.

8. Current state per game (measured, not claimed)

- FFVI old corpus: 117 fires, overlap 76/83, missed 7, extra 41, NE 0.36, latency budget OK. FFVI new corpus: 104 fires, overlap 46/50, missed 4, extra 58. Zelda long: 71-72 fires, overlap 50/60, missed 10, extra 21-22, panellat EXCEEDED pre-existing (identical stash baseline — not a regression). Zelda intro: 4 growth fires, overlap 0 (fast-scroll pages outrun stability). Paper Mario clean run: 485 reads, 28 fires all intro, sim reproduces 28/28 with zero extras; gameplay silence = sidecar deaths plus focus holds plus static skips (diagnosed, not policy). Dims banked per session (PM 788x583). Exact-match overlap understates under variant churn (proven: clean-form recovery scores as miss plus extra).

9. Truth eval set v1 (Step 1, 7e4977a)

- eval/truth.jsonl: 40 rows from 36 banked shots, game pane read directly (never OCR as ground truth). Dev 28 (17/18h sessions, tuning allowed), heldout 12 (22/23h, verification only). Verdicts: 17 miss-complete, 6 false-pub, 4 fidelity-error, 5 correct-silence controls, 2 hits, plus stale/dup/menu cases. Contexts span dialogue, typewriter partials (labeled by visible text), crawl pages, signs, 2 decision-box misses, game-over/menu, JA and Latin titles, prompt-leak garbage, ?-junk, blends, dups. Initial set only — no Paper Mario yet, not cross-game proof. Data-only batch. Revert: revert truth-set.

10. Failure taxonomy and residue inventory

- Recognition (OCR misreads; beam N-best available but discarded until Step 2 banking), candidate formation (singletons never form — Tier 1/2 scope), commit policy (stability vs completeness; singleton residue ~35 FFVI-new + ~39 Zelda-long + 3 intro), timing (weak-hold 16s p50 Zelda; intro outrun), segmentation (ケアルファイア merge), temporal association (moving text, blends), false positives (menu garbage, prompt leak, ?-junk, notification hallucinations), duplicates (coal-mine double post, repeat fires), context (pack-less PM, unbanked names), downstream (translation follows JA errors: 嫁 soldiers, Quarfia). Title failures: sword-scene mistitle (vetoed), fast-skipped titles, 5-minute-late TITLE. Mash behavior: policy question open (keep stale / fast-track / expire — recommended expire; no build until answered).

11. Dictionary stance and Step-3 experiment (agreed, unbuilt)

Position: premade package yes (SudachiPy first choice for measurement; frozen wordlist for integration; license verified before shipping), re-rank N-best never rewrite, visual evidence primary, pack/memo names exempt, incomplete shapes exempt, frequency never overrules visual support. OCR verified vision-only (no LM, no dictionary); beam strings exist and are now banked.

Accepted experiment: truth set with dev/heldout split; four questions (candidate presence, re-rank wins, damage split by names/unusual/partial, CER plus exact-line plus latency); scoring frozen before measuring (beam-rank dominant, lexicon breaks near-ties, clear margin or no change; correct-reading-absent logged as generation limit); acceptance is corrections >=3x regressions with zero name damage, p95 +<=50ms/line, candidate-absent rate reported; fail means keep beam-diversity flagging plus fixup mining. Integration is a separate reversible batch only on pass, with corpora re-runs and cross-game checks.

12. Title, sprite, and decision-box policy (all agreed, mostly unbuilt)

- Logo posts TITLE; press-start prompt posts as in-game text right after; English credits never post (audited: zero Latin-only publications). Zelda sprites: heart to <3 inline, sign arrows <- / down-arrow inline, Link-face ignored, brackets 1:1, leading-glyph junk treated as sprite misreads. Decision boxes (Zelda arrow, PM pointing hand): positioned question plus options in exact in-game positions, pointer sprite never posted. Sprite/positioned-box builds are queued behind Paper Mario data (Step 10).

13. Milestone path

Done (all gated, all committed)

- Steps 1-2 (truth set, beam banking). Tiers 1-2 (fuzzy-window, anchored-singleton). Loops 1-5 (global fixups, confusion re-mine, memo persist plus Stop closeout, vocab proposals, profiles). Title hardening, ROM-ID identity plus paren fix, engine notice, death forensics, PM pack v1, shot labeling, capture-settings and emulator-version policies.

Planned in order

- Step 3 dictionary experiment (needs truth plus beams: both banked). Steps 4-6 Zelda/FFVI fixes driven by triage (title live proofs, missed-line sweep, LM-fluency gate only on evidence). Steps 7-8 Paper Mario baseline play then pack cross-game check (promote only what generalizes). Step 9 loop payoffs (re-mine and promotion review with PM data; profile tuning only on measured spread). Step 10 big queued builds in evidence order (decision boxes, sprites, shop UI on user signal). Never a rigid chain blocking onboarding: baseline play proceeds in parallel.

14. Open questions for the user (nothing proceeds on these without answers)

- Mash policy a/b/c (keep stale / fast-track / expire to empty). Shop-play signal (shop UI waits). Wedge/Biggs official English (user call). EN glossary confirmations for the PM pack. Which of death-instrumentation findings vs PM pack vs dictionary experiment goes next after this handover.

15. Live proofs owed (confirm in the next play triage)

- [memo] saved/loaded lines. Undoubled ROM-ID plus normalized line on Dolphin. TITLE-then-prompt on Paper Mario; mid-game TITLE silence. Engine notice and forensics paths (need a real 4-death session — hopefully never). PM pack routing, banking IDs, TITLE on the PM logo. Beam content in new corpus lines (banking already proven once: 314/343).

16. Working conventions (binding)

- Run protocol (user plays, exact instructions plus keeper state, single confirmatory runs, solo reads, taskkill-first on red). Gates (replay drift 0, simreplay parity plus JA-set diffs, panellat budgets, build +0/0). Revert phrase per batch (ledger above). Comms short and direct, text-only except geometry/truth shots. Objectivity over agreement; investigate before confirming beliefs. Verify own solutions by execution.

17. File map and key pointers

- Host: Capture/CaptureController.cs (pipeline, stability, commit roads, miner, memo, profiles, singleton/fuzzy branches), Capture/SimReplay.cs (filename pack routing, overlap/missed/extra reports), Packs/ (IPack contract, PackRegistry routing and content votes, per-game packs incl. PaperMario/), Sidecar/ (process manager with budget/notice/forensics, job object), Ocr/OcrClient.cs (lenient parse incl. beams), Translation/ (DeepL/CAT/cache/gates), Windows/ (panel history, menu/battle frames, shot saver with session labels). Sidecar: ocr-service/main.py (detection, beamed recognition, Div, beams banking). Tools: mine-confusion.py (loop 2), build-handover-pdf.py (this document). Evidence: eval/truth.jsonl (README documents split/verdicts), BOOKS.md (full ledger with commits and reverts), PERFORMANCE.md (SLOs). Corpus/sidecars/caches: never commit translation_cache.json (checkout after replays).

18. Known risks and watch items

- Sidecar mid-session deaths (4x in the PM clean run; cause class still unproven — forensics will name it). One crash-truncated tail gzip in the corpus dir (not blocking). One orphaned PM miner file under the pre-paren versioned id (negligible). Zelda-long latency budget exceedance is pre-existing, not a regression. Exact-match metrics understate variant correctness (accounted in verdicts, not in counters). Sim and replay never touch bitmaps, the loop, or the quiet-title path — vision and title changes are live-provable only. Static-frame skip plus focus holds explain most gameplay silence; distinguish them from policy before building.